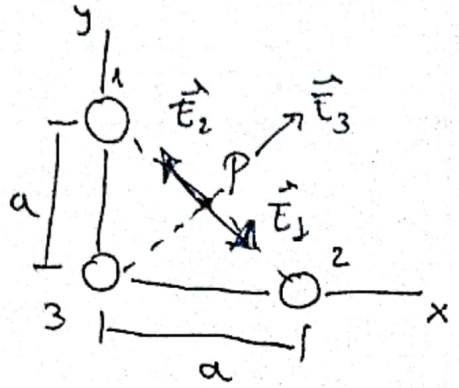
Gabarito da 3ª VA de FG3 – 27/04/23

Prof. Fábio Novaes

1)

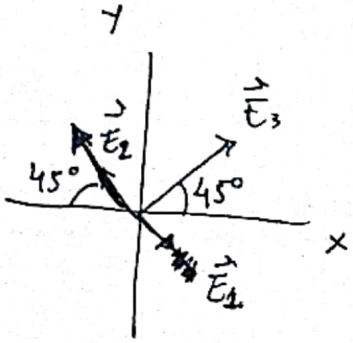


$q_1 = e$ $\quad a = 2{,}0 \times 10^{-6}\,m$

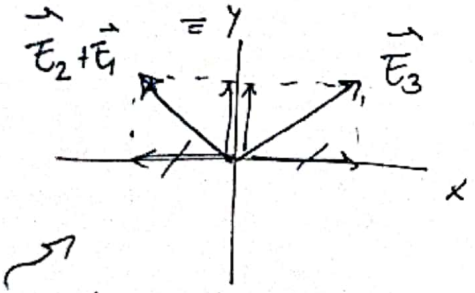
$q_2 = 3e$

$q_3 = 2e$

▶ Campo elétrico resultante em P:



$$\begin{cases} E_1 = k\dfrac{e}{d^2} \\ E_2 = k\dfrac{3e}{d^2} \\ E_3 = k\dfrac{2e}{d^2} \end{cases}$$



$\vec{E}_2+\vec{E}_1$ e $\vec{E}_3$ têm mesmo módulo.

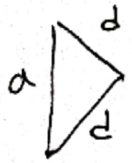
$$|\vec{E}_2+\vec{E}_1| = E_2 - E_1 = k\frac{2e}{d^2} = E_3.$$

$$E_{P,x} = 0$$

$$E_{P,y} = 2E_3 \operatorname{sen} 45° = E_3\sqrt{2}$$

$$= k\frac{2e}{d^2}\sqrt{2}$$

Achando d:



$$2d^2 = a^2$$

$$d = \frac{a}{\sqrt{2}} \Rightarrow E_{P,y} = k\frac{4e\sqrt{2}}{a^2} = 8{,}99\times10^9 \cdot \frac{4\times1{,}6\times10^{-19}\sqrt{2}}{4\times10^{-12}}$$

$$= 8{,}99\cdot1{,}6\cdot\sqrt{2}\times10^2$$

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$$\therefore \boxed{E_{P,y} = 20{,}34 \times 10^2 \approx \underbrace{2000\ N/C = 2\times10^3\ N/C}_{\text{2 algarismos significativos}}}$$

$$\boxed{\vec{E}_P = E_{P,y}\,\hat{j}}$$

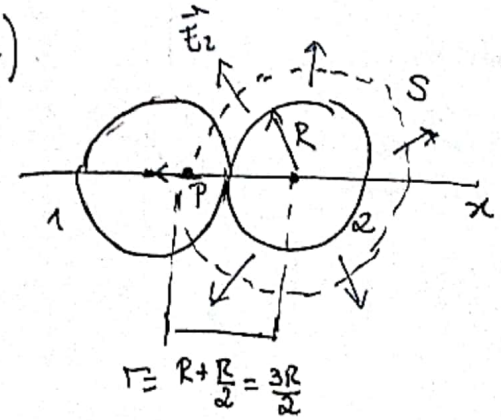
b) $q = -e$

$$\vec{F}_P = -e\vec{E}_P = -1{,}6\times10^{-19}\cdot 20{,}3\times10^2\,\hat{j}$$

$$= -32{,}5\times10^{-17}\,\hat{j}$$

$$\boxed{\vec{F}_P = -3{,}3\times10^{-16}\,N\,\hat{j}}$$

2) a)



S: superfície Gaussiana

$$\oint_S \vec{E}_2 \cdot d\vec{A} = \frac{q_2}{\varepsilon_0}$$

$$\oint_S E_2\, dA = \quad ; \quad \vec{E}_2 \parallel d\vec{A}$$

$$E_2 A_S = \frac{q_2}{\varepsilon_0}$$; E_2 só depende do raio, por simetria.

$$\therefore \boxed{E_2 = \frac{q_2}{4\pi\varepsilon_0 r^2} \overset{r=\frac{3R}{2}}{=} \frac{q_2}{\cancel{4}\pi\varepsilon_0 \frac{9R^2}{\cancel{4}}} = \frac{q_2}{9\pi\varepsilon_0 R^2}} \quad (1)$$

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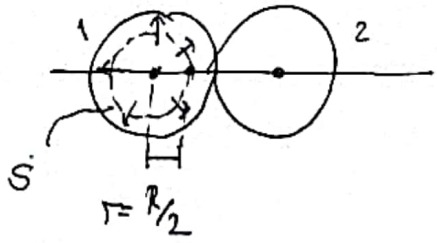
b)



$$\oint_S \vec{E}_1 \cdot d\vec{A} = \frac{q_{env}}{\varepsilon_0}$$

$$E_1 A = \frac{q_{env}}{\varepsilon_0}$$, por simetria.

$$E_1 = \frac{q_{env}}{4\pi\varepsilon_0 r^2} = \frac{q_{env}}{\cancel{4}\pi\varepsilon_0 \frac{R^2}{\cancel{4}}} = \frac{q_{env}}{\pi\varepsilon_0 R^2} \quad (2)$$

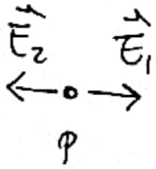
Achando q_{env}:

$$\rho = \frac{q_1}{\frac{4\pi}{3}R^3} = \frac{q_{env}}{\frac{4\pi}{3}r^3} \Rightarrow q_{env} = q_1 \frac{r^3}{R^3}$$

$$\xrightarrow{r=\frac{R}{2}} q_{env} = \frac{q_1}{8} \quad (3)$$

$$(3) \rightarrow (2) \Rightarrow \boxed{E_1 = \frac{q_1}{8\pi\varepsilon_0 R^2}} \quad (4)$$

c) $\vec{E}_P = \vec{E}_1 + \vec{E}_2 = (E_1 - E_2)\hat{\imath} \overset{!}{=} 0 \Rightarrow E_1 = E_2$



$$\overset{(1)e(4)}{\Rightarrow} \frac{q_1}{8\pi\cancel{\varepsilon_0}R^2} = \frac{q_2}{9\pi\cancel{\varepsilon_0}R^2}$$

$$\therefore \boxed{\frac{q_2}{q_1} = \frac{9}{8}}$$

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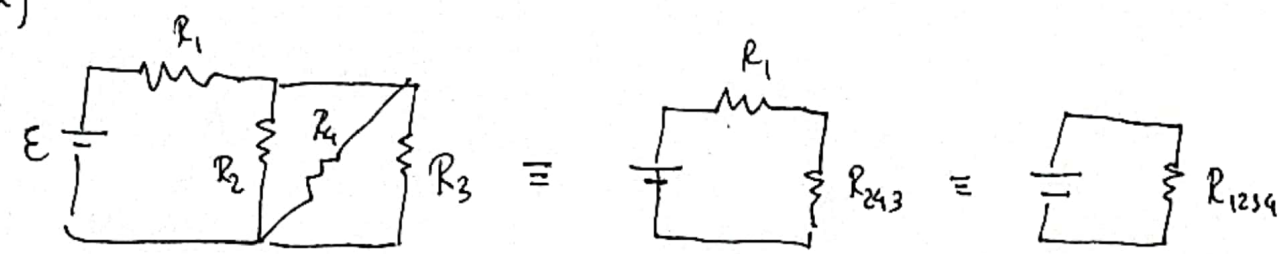
3) $R_1 = 4\,\Omega$, $R_2 = R_3 = 20\,\Omega$, $\mathcal{E} = 10\,V$

a)



2,4 e 3 em paralelo

1 e 243 em série

$$\rightarrow \frac{1}{R_{243}} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \underset{R_2 = R_3}{=} \frac{2}{R_2} + \frac{1}{R_4} \underset{R_2 = 20\Omega}{=} \frac{1}{10} + \frac{1}{R_4} = \frac{R_4 + 10}{10 R_4}$$

$$\therefore R_{243} = \frac{10 R_4}{R_4 + 10} \quad (1)$$

$$\boxed{R_{1234} = R_1 + R_{243} \overset{(1)}{=} 4 + \frac{10 R_4}{R_4 + 10} = \frac{14 R_4 + 40}{R_4 + 10}} \quad (2)$$

era pra ser 25 Ω!

b) $P_{fonte} = \mathcal{E}\, i = 10\, i = 50 \Rightarrow i = 5A$ (3)

2,5A

Lei das malhas: $\mathcal{E} = R_{1234} \times i$

$$10 \overset{(2)\,e\,(3)}{=} 5 \left(\frac{14 R_4 + 40}{R_4 + 10} \right)$$

$$2 = \frac{14 R_4 + 40}{R_4 + 10}$$

$$2R_4 + 20 = 14 R_4 + 40$$

$$12 R_4 = -20 \Rightarrow R_4 = -\frac{5}{3}\,\Omega = -1{,}66\,\Omega$$

$$\Rightarrow \boxed{R_4 = 0}$$

Era pra ser zero.

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c)

$$P = 10\,i\,, \qquad i = \frac{\mathcal{E}}{R_{1234}} = \frac{(R_4+10)}{14R_4+40}\times 10$$

$$\therefore\ P = 100\left(\frac{R_4+10}{14R_4+40}\right) \xrightarrow{R_4\to\infty} \frac{100\,\cancel{R_4}}{14\,\cancel{R_4}} = \frac{100}{14} = \frac{50}{7}$$

$$\boxed{P_\infty = 7{,}1\ W}$$

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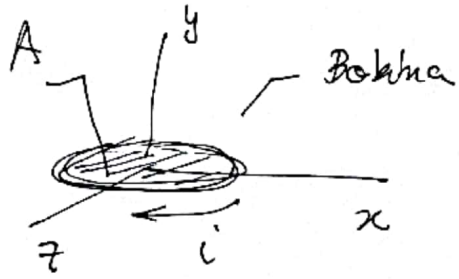
4) $i = 3\ A$

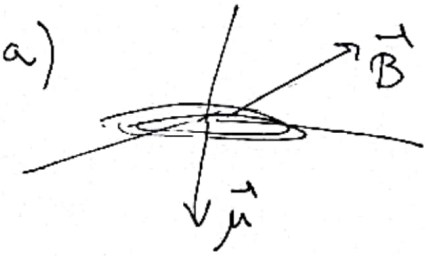
$N = 3$

$A = 6{,}0\times 10^{-3}\ m^2$

$\vec{B} = (3\hat{i} + 2\hat{j} - 4\hat{k})\times 10^{-3}\ T$



a)



$$U_B = -\vec{\mu}\cdot\vec{B}$$

$$\vec{\mu} = -NiA\,\hat{j}$$

Pela regra da mão direita

$\vec{\mu} = -|\mu|\hat{j}$

$$= -3\cdot 3\cdot 6\times 10^{-3}\,\hat{j}$$

$$\vec{\mu} = -54\times 10^{-3}\,\hat{j}$$

$$\therefore\ U_B = -|\mu|\,\underbrace{\hat{j}\cdot(3\hat{i}+2\hat{j}-4\hat{k})}_{=0}\times 10^{-3}\ ; \quad \left(\hat{j}\cdot\hat{i} = 0 = \hat{j}\cdot\hat{k}\ ;\ \hat{j}\cdot\hat{j} = 1\right)$$

$$= -54\times 10^{-3}\cdot 2\times 10^{-3} = -108\times 10^{-6}\ J$$

$$\boxed{U_B = -0{,}11\ mJ}$$

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b) $\vec{\tau}_B = \vec{\mu} \times \vec{B}$

$$= -|\mu|\ \hat{j} \times (3\hat{i} + 2\hat{j} - 4\hat{k}) \times 10^{-3}$$

$$= -|\mu| \left[3\underbrace{(\hat{j}\times\hat{i})}_{=-\hat{k}} + 2\overbrace{(\hat{j}\times\hat{j})}^{=0} - 4\underbrace{(\hat{j}\times\hat{k})}_{=\hat{i}} \right] \times 10^{-3}$$

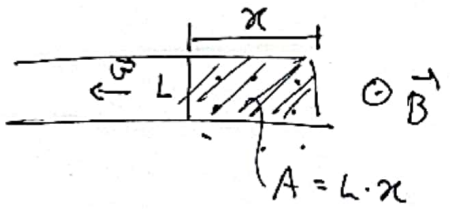
$$= -|\mu| (-3\hat{k} - 4\hat{i}) \times 10^{-3}$$

$$= 54 \times 10^{-6} (4\hat{i} + 3\hat{k})$$

$$= 216 \times 10^{-6}\ \hat{i} + 162 \times 10^{-6}\ \hat{k}$$

$$\boxed{\vec{\tau}_B = (0{,}216\ \text{mN}\cdot\text{m})\hat{i} + (0{,}162\ \text{mN}\cdot\text{m})\hat{k}}$$

5) $B = 0{,}50\ T$



a) $L = 20{,}0$ cm

$v = 60$ cm/s

$\mathcal{E} = ?$

Lei de Faraday: $\mathcal{E} = -\dfrac{d\Phi_B}{dt}$

$$|\mathcal{E}| = \frac{d\Phi_B}{dt}, \quad \Phi_B = \int_{\text{Região } L\times x} \vec{B}\cdot d\vec{A}$$

$$\Phi_B = BA = BLx$$

$$\therefore\ |\mathcal{E}| = BL\frac{dx}{dt} = BLv$$

$$\boxed{|\mathcal{E}| = 0{,}5 \cdot 0{,}2 \cdot 0{,}6 = 60 \times 10^{-3} = 60\ \text{mV}}$$

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b) $R = 20{,}0\,\Omega$

Lei do Ohm: $\mathcal{E} = Ri \Rightarrow i = \frac{\mathcal{E}}{R} = \frac{6\cancel{0} \times 10^{-3}}{2\cancel{0}}$

$$\boxed{i = 3\,mA}$$

c) $P = Ri^2 = 20 \cdot (3 \times 10^{-3})^2 = 20 \cdot 9 \times 10^{-6}$

$= 180 \times 10^{-6}$

$$\boxed{P = 0{,}18\,mW}$$

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